

HealthConnect Clinic Experience Lab

Week 7 Technical Report & Data Quality Audit Findings

Project Scope: Evaluation of appointment records, operational timing distributions, categorical consistency, and dashboard architectural deployment via Python/Streamlit.

1. Executive Summary & Architecture Overview

During the Week 7 cycle of the HealthConnect Clinic Experience Lab, rigorous data validation protocols were executed against the dataset `healthconnect_appointment_data.csv`. The implementation leverages a local Linux environment paired with Python, Pandas, and Plotly to establish a reliable operational dashboard infrastructure.

All structured validation checks—including primary key uniqueness, strict zero-duplicate threshold scans, and dictionary-aligned missing value isolation—have successfully completed with 100% verification.

2. Final Key Performance Indicators (KPIs)

Category	KPI / Validation Metric	Confirmed Value	Status
Data Integrity	Primary Key Uniqueness (appointment_id)	100% Unique (Zero Collisions)	PASSED
Data Completeness	Duplicate Record Scan (DataFrame level)	0 Duplicate Rows Found	PASSED
Operational Control	Missing Value Isolation	Restricted to Designated Fields	PASSED
Categorical Quality	Consistency (Outcomes & Types)	Zero Rogue Strings / Formatting Faults	PASSED
Dashboard System	Dynamic Script Integration	Synchronized with Auto-Detection	PASSED

3. Environment & Deployment Summary

Environment Configuration

Managed via Python virtual environment (venv) under Linux architecture to circumvent PEP 668 external-package blocks, ensuring safe and isolated dependency rendering.

Data Pipeline

Dynamic fallback handling and glob-based file path resolution guarantee seamless loading of `healthconnect_appointment_data.csv` directly inside the application workspace.